

# **CTSpinoPelvic1K: spine, pelvis, ribs and femora in one coordinate frame, annotated for lumbosacral transitional anatomy**

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### **Author contribution statement**

G.S. designed the dataset and the label scheme, implemented the processing pipeline and the quality-control procedures, and drafted the manuscript. A.S., A.T., M.K., R.C., D.H., F.M., H.S., M.S., S.S., M.M. and J.K. performed annotation correction and quality-control review of the released segmentations under supervision. M.I. and N.A. performed the radiological reads — the Castellvi grading and the adjudication of candidate instrumentation. [BEFORE SUBMISSION: confirm each contribution individually with the author concerned and expand to CRediT roles.] All authors read and approved the final version of the manuscript.

### **Conflict of interest statement**

The authors have no relevant conflicts of interest to disclose.

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### **Data and code availability (unblinded)**

Dataset, archived before submission: <https://doi.org/10.5281/zenodo.22139642> (concept identifier, resolves to the current version); the version described in the manuscript is v7, <https://doi.org/10.5281/zenodo.22242745>.

Code: <https://github.com/Gregory-Schwing-MD-PhD/CTSpinoPelvic1K>, tagged at the release commit.

Mirror: <https://huggingface.co/datasets/OpenSpineConsortium/CTSpinoPelvic1K> — a convenience mirror, not the archive of record.

### **Ethics**

This work uses publicly available, de-identified imaging and annotations derived from it. A determination of not-human-participant research has been submitted to the Wayne State University Institutional Review Board.

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[Author names removed for double-anonymized review]

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**Purpose:** Lumbar levels cannot be numbered reliably when a lumbosacral transitional vertebra (LSTV) is present. Levels are conventionally established by counting downward from C2, the reference standard for vertebral numeration<sup>1</sup>. An abdominal field of view lacks that landmark, leaving a specific ambiguity: four rib-free lumbar vertebrae may mean an L1 bearing a lumbar rib or an L5 assimilated to the sacrum, and six may mean a true sixth lumbar vertebra or a T12 whose ribs are aplastic<sup>2</sup>. CTSpinoPelvic1K is built to test whether local morphology can resolve them without counting from the C2 anchor. It provides 802 CT records anchoring each lumbar level on the lowest rib-bearing vertebra above and the first sacral segment below, with radiologist-sourced spine *and* pelvic annotation on one coordinate frame, per-level ribs and femora. Classes for a sixth lumbar vertebra, a thirteenth thoracic vertebra, a separate first sacral segment, and lumbar ribs record a variant as itself. For lumbosacral surgical planning and level-numbering research.

**Acquisition and Validation Methods:** Records pair TCIA CT COLONOGRAPHY imaging with CTSpine1K<sup>3</sup> vertebral and CTPelvic1K<sup>4</sup> pelvic labels, each placed on the acquisition with highest bone coverage, unified under a VerSe-native scheme. Validation: geometric invariants (802/802 pass); rib-vertebra incidence across 5,749 evaluable ribs (2 offsets, 0.035%); and agreement of derived spinopelvic measures with published values.

**Data Format and Usage Notes:** Gzipped NIfTI image/label pairs sharing one affine, canonicalized to PIR, with frozen patient-grouped LSTV-stratified five-fold splits and a reference loader. Archived at 10.5281/zenodo.22139642.

**Potential Applications:** Segmentation and level-numbering benchmarks, variant studies, surgical planning, opportunistic screening. *Limitations:* thoracic ground truth is field-of-view limited; postural angles are supine; splits include no held-out test set; the rib layer is a pseudolabel whose review was triaged, not exhaustive; and the Castellvi grades were assigned by a radiology resident physician.

## 6 I. INTRODUCTION

7 Labeling a lumbar vertebra requires counting, and counting requires two fixed ends.  
8 The spine commonly carries seven cervical, twelve thoracic and five lumbar vertebrae, but  
9 *enumeration anomalies* — eleven or thirteen thoracic, four or six lumbar — are common, and  
10 the transitional phenotypes that produce them sit at the two ends of the lumbar column:  
11 the lumbosacral transitional vertebra (LSTV) below and the thoracolumbar transitional  
12 vertebra (TLTV) above. In the presence of an LSTV the lower end is not where it is  
13 assumed to be. The two phenotypes are distinct developmental events — a lumbar segment  
14 assimilating to the sacrum, or a sacral segment separating from it — and they leave different  
15 traces: a sacralized L5 tends toward an anteriorly wedged body, a reduced disc beneath it and  
16 hypoplastic or absent facet joints, while a lumbarized S1 tends toward a squared body in the  
17 sagittal plane, well formed lumbar-type facets and a full-height disc<sup>2</sup>. What counting settles  
18 is the *name*. The difficulty is that these traces are not decisive in every case, so the name and  
19 the morphology can disagree, and the count that would arbitrate is taken downward from  
20 C2 — unavailable in any spine-limited acquisition. LSTV is common, reported between 4%  
21 and 30% depending on definition, and wrong-level surgery is its most serious consequence.

22 Existing public collections do not make this tractable, and not because they are small  
23 (Fig. 1): the spine collections omit the pelvis, the pelvic ones carry no vertebral count, and  
24 the whole-body schemes have neither an L6 class nor a class for a rib borne by a lumbar  
25 vertebra, so a six-lumbar spine cannot be written down in them at any segmentation quality.  
26 VerSe<sup>5,6</sup> is the instructive case, and Fig. 1(b) gives it: a collection can enrich for anatomical  
27 variants deliberately, grade them, and still be unable to represent the ones it grades.

28 CTSpinoPelvic1K places the spine, pelvis, ribs and femora in one coordinate frame and  
29 gives a class to each structure an enumeration anomaly produces: a sixth lumbar vertebra,  
30 and a rib borne by a lumbar vertebra. A scheme without those classes must record such  
31 anatomy as something it is not.

32 *Where the material is.* Everything described here is v7 of a single archived dataset,  
33 deposited at <https://doi.org/10.5281/zenodo.22242745>. Citations should use <https://doi.org/10.5281/zenodo.22139642>, the permanent identifier for the dataset across all  
34 of its versions, which resolves to whichever version is current; the version identifier above  
35 pins the exact release measured in this manuscript. Documentation, the label scheme, the

reference loader and the per-record quality-control tables are distributed with the archive  
and are described in Sec. [III](#).

### **I.A. Vertebra labeling when the scan cannot settle the count**

*The limitation, stated plainly.* No scan in this corpus contains C2 or the whole cervical spine, so the conventional count cannot be performed. That is a property of abdominal imaging rather than of the annotation, and it means a variant at the thoracolumbar junction cannot be *resolved by counting* here at all: a thirteenth thoracic vertebra, a lumbar rib, and a *stump rib* — a hypoplastic twelfth rib, one of the cardinal indicators of a thoracolumbar transitional vertebra — produce overlapping appearances.

*What the release does instead.* No vertebral label is asserted where it cannot be supported. Every quantity is expressed against the two landmarks present in essentially every abdominal scan — the sacrum below and the lowest rib-bearing vertebra above — so the description is positional rather than nominal: a vertebra by how many rib-free vertebrae separate it from the sacrum, a lowest rib by its length as a fraction of the rib above it, a transverse process by its span and its distance from the ala. None of these changes according to whether a reader would have said L5 or S1, so cases that disagree about the name remain comparable. Where the junction is in view the count is determined and the conventional name follows, and the two descriptions agree.

*And the morphology may be local rather than global.* Counting is not the only route to an identity. A thoracic vertebra differs in shape from a lumbar one — in the span of its transverse processes relative to its body, in the proportions of its endplates and canal — and if that difference holds at the boundary itself, the phenotype is decidable from the vertebra in front of the reader without any global count. On this corpus it does hold: separating T12 from L1 on shape alone, with each patient’s own size divided out and cross-validation grouped by case, gives an area under the curve of 0.990 and 97.3% accuracy over 1485 vertebrae. Size is deliberately removed, because a classifier given raw millimeters separates thoracic from lumbar immediately and has learned nothing that helps at the junction, where the two are nearly the same size.

That is reported as a property of the released measurements rather than as a method: the information needed to settle these variants is present locally.

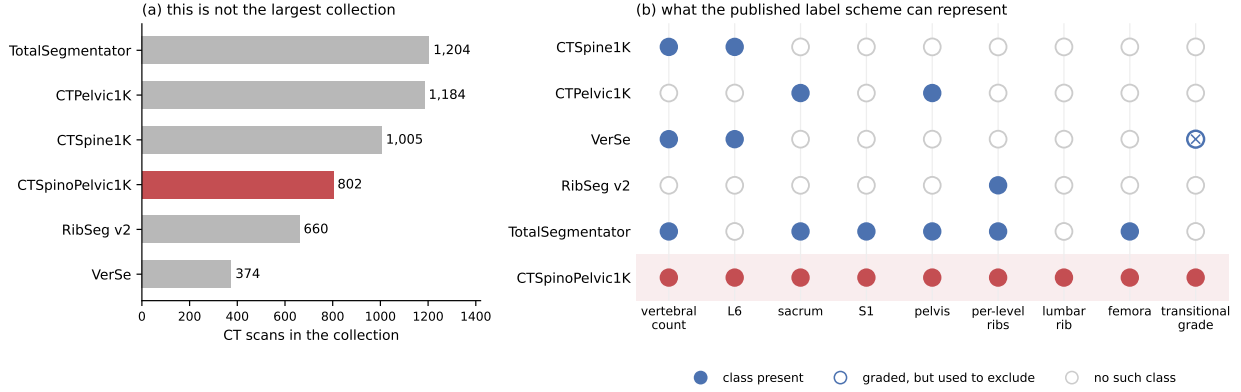


FIG. 1 Public CT collections at the lumbosacral junction. (a) Size, stated first because it is the claim this dataset does *not* make: three of the five comparators carry more scans, and RibSeg v2<sup>7</sup> labels more ribs. (b) What each published scheme can *represent*, a property of its class list rather than of its segmentation quality. VerSe is graded-but-excluding: it applies the Castellvi classification and uses it to decide what to omit, leaving vertebrae partially fused to the sacrum unsegmented<sup>6</sup> — 25 of the 33 graded records here.

## II. ACQUISITION AND VALIDATION METHODS

### II.A. Sources, and why a crosswalk was necessary

Both CTSpine1K<sup>3</sup> and CTPelvic1K<sup>4</sup> draw part of their imaging from the TCIA CT colonography collection<sup>8,9</sup>, and the vertebral and pelvic annotations in both were produced under board-certified radiologist supervision. Neither released the mapping from an annotation to the specific CT series it was drawn on.

That mapping is not a bookkeeping detail. Every patient in CT COLONOGRAPHY was scanned *twice*, prone and supine, often with more than one kernel, so one patient identifier resolves to several volumes differing in posture, position, field of view and noise — and an annotation carries a patient identifier and nothing finer.

Placing it on the wrong volume of the same patient is not merely a displacement. The spine is mildly deformable, and turning a patient from supine to prone alters lumbar lordosis and the alignment of each segment against the next, so a vertebra moves by a different amount, and in a different direction, from the one above it. Outlines drawn on one series are the wrong *shape* for the other, and no rigid realignment recovers them. In isolation such



82 a mask looks like competent work; it reveals itself only when overlaid on the image.

83 The crosswalk (Fig. S1) therefore resolves each annotation first to a patient and then,  
84 within that patient’s volumes only, to the series whose bone *agrees* with it: candidates are  
85 thresholded at bone attenuation and scored by overlap with the annotation mask. Bone  
86 rather than image similarity, because bone is what the annotation describes — two acqui-  
87 sitions of one abdomen resemble each other everywhere except in where the skeleton sits.

## 88 II.B. Fused and split records

89 The two annotation sets were placed independently, so they do not always land on the  
90 same series. Of 802 records, 342 carry spine and pelvic annotations on the same volume  
91 (*fused*), 351 are *separate*, 89 are *spine\_only* with no pelvic annotation for that patient, and  
92 20 are *pelvic\_only*. These four are the values of the manifest’s `match_type` field; the coarser  
93 `config` field folds *separate* into *spine\_only*, so a user wanting the four types should read  
94 `match_type`. The split is the crosswalk’s main finding rather than bookkeeping: in those  
95 separate records — nearly half of all patients — the two public collections annotated *different*  
96 *volumes of the same person*, and neither recorded which. A pelvic annotation placed on the  
97 other acquisition of a prone/supine pair cannot be combined with the spine annotation  
98 without resampling across a change of posture. Those records are retained rather than  
99 discarded, and held out of the postural analyses, because two acquisitions of one patient  
100 differing in spinal alignment are a resource for a mobility study rather than a defect.

## 101 II.C. Densification, and why it is asymmetric

102 A corpus in which half the records lack a pelvis is not usable as a spinopelvic dataset,  
103 so the missing structures were pseudolabeled — asymmetrically (Fig. S1), because the two  
104 directions are not equivalent.

105 Pelvis onto spine-only records is the safe direction. The sacrum and innominate bones  
106 are single objects with no enumeration: no count to get wrong, no naming convention to  
107 choose, no analog of a transitional vertebra. Quality is reported as held-out performance on  
108 the *pelvic\_only* records, withheld from training for exactly this purpose.

109 The reverse direction carries the whole problem this dataset exists to address, since a

110 pseudolabeled spine must commit to a vertebral count and on a transitional vertebra will  
111 commit to the commoner reading. Those 20 records were therefore corrected by hand rather  
112 than densified automatically — tractable at twenty cases, which is the practical reason the  
113 asymmetry runs the way it does.

## 114 II.D. Ribs

115 No public rib annotation covers this cohort, and the two available automatic sources fail  
116 in complementary ways.

117 Möller’s binary rib network<sup>10</sup> is a detection model of high reported accuracy, but it  
118 segments ribs that lie *entirely within* the field of view; a rib clipped by the edge of an  
119 abdominal acquisition is liable to be missed altogether. Since this is a colonography cohort,  
120 the lower ribs are routinely clipped, and those are precisely the ribs the count depends on.

121 TotalSegmentator<sup>11</sup> does segment clipped ribs and supplies the per-rib numbering that a  
122 binary network cannot. Its characteristic failure is at the other end of the bone: the most  
123 medial portion of the rib — roughly the last tenth to sixth approaching the costovertebral  
124 joint — is frequently absent. A rib truncated at its medial end never reaches the vertebra  
125 it articulates with, so the incidence test that assigns a rib to its level has nothing to test.

126 The two were therefore unioned (Fig. S1): TotalSegmentator supplies numbering and the  
127 medial-clipped cases, Möller supplies detection where its output is more complete, and the  
128 union carries TotalSegmentator’s per-level identities.

129 The result is a pseudolabel whose review was triaged rather than exhaustive, which is  
130 worth stating plainly because a layer described only as “human-corrected” invites the reader  
131 to assume every bone was looked at. A label-based rule flagged every record in which one  
132 rib bone carried more than one identifier, or in which a rib failed to reach the vertebra it  
133 articulates with — a single bone with two numbers being a numbering error by construction,  
134 whatever the anatomy. The rule selected 152 records; 148 were reviewed and corrected, and 4  
135 were not and are identified in the release. Reviewers were medical students working through  
136 OpenSpineConsortium<sup>12</sup>, trained against a written labeling protocol, with every correction  
137 recorded against the annotator who made it. Records passing the rule were not individually  
138 inspected, so the layer’s guarantee is that it is free of the failure modes the rule detects,  
139 and no stronger. The residual rib–vertebra offsets reported in Sec. II.G are the independent

140 check on what the rule missed.

## 141 II.E. Label scheme and counting anchors

142 The scheme is VerSe-native: vertebrae keep their VerSe identifiers (C1–C7 = 1–7, T1–T12  
143 = 8–19, L1–L6 = 20–25, sacrum 26), and every non-VerSe structure takes a fixed identifier  
144 above that range — S1 carved from the sacrum (29), hips (30–31), femora (32–33), ribs  
145 (34–45 left, 46–57 right), soft tissue (58–73), lumbar ribs (74–75) and surgical hardware  
146 (76–82).

147 Two anchors make the count decidable, and Fig. 2 shows both: the lowest rib-bearing  
148 vertebra above, **S1** carved from the sacrum below. Each earns its place by what it supports  
149 rather than by being an endpoint — S1 supplies the sacral endplate landmark that sacral  
150 slope and pelvic incidence are measured from, and with both brackets present L5 and L6  
151 are distinguishable where a spine-limited view alone would not be.

152 *How S1 is cut.* The sacrum’s outer boundary is the source annotation and is never al-  
153 tered; S1 is the cranial part of that same bone, separated by a plane. The plane’s normal  
154 is the sacrum’s own cranio-caudal axis, made orthogonal to a left–right axis that is mea-  
155 sured rather than assumed — the normal of the reflection plane best mapping the sacrum  
156 onto itself, which removes the patient’s rotation within the scanner. That rotation is not  
157 negligible: median 4.5°, maximum 16.1°, and 508 of 801 records above 3°. The level of the  
158 cut preserves the sub-division volume of the preceding release, so the plane changes the  
159 shape and orientation of the S1/S2 boundary and not how much of the sacrum is called S1.  
160 Per-record geometry ships with the archive.

161 Neither anchor is novel: TotalSegmentator<sup>11</sup> carries a separate S1 class and per-level  
162 ribs, and VerSe<sup>5</sup> carries L6. Because this release uses the VerSe identifiers, its L6 *is* VerSe’s,  
163 interoperable rather than local to this work. L6 is retained not for novelty but because a  
164 scheme without it cannot record a six-lumbar spine at all, and must renumber the column  
165 or drop a level to fit.

166 **A rib on a lumbar body receives its own class**, and this is the one class here with no  
167 published counterpart. A lumbar rib and a stump rib are the same object under two counts,  
168 so assigning it a number would commit the label to one reading of an enumeration anomaly.  
169 A scheme that numbers every rib 1–12 has nowhere to put a thirteenth: the annotator must

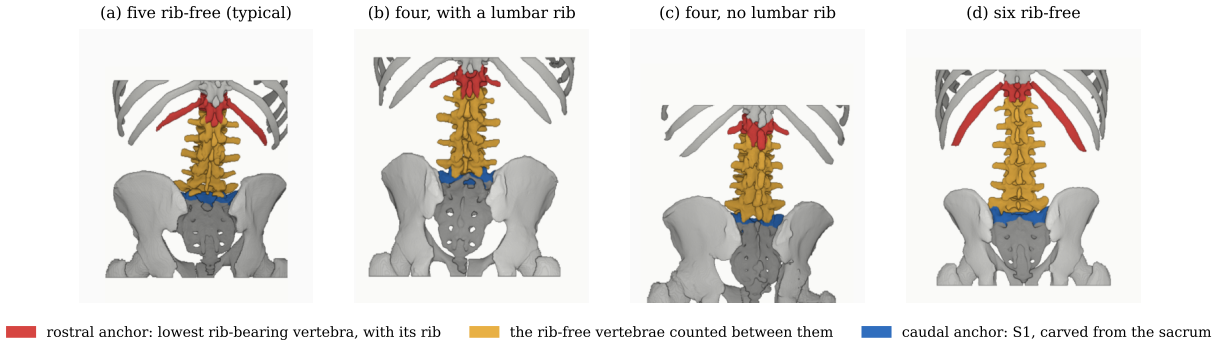


FIG. 2 The two anchors and the interval between them, from the released labels, posterior view. Rostral anchor: the lowest rib-bearing vertebra and the rib that makes it one. Caudal anchor: S1. Both are found from the labels rather than assumed, and panels are colored by role rather than identifier, since coloring T12 would assert the count the figure derives. Panels (b) and (c) reach the same interval of four from opposite ends and are indistinguishable by it. All panels are at one millimeter scale, aligned at the sacral base.

170 either call it rib 12 — asserting the vertebra beneath it is thoracic, the very question at  
 171 issue — or discard it. TotalSegmentator carries ribs 1–12 per side and no such class, and  
 172 VerSe’s T13 is a vertebra rather than a rib. 16 released records carry a lumbar rib and 18  
 173 carry an L6.

174 Figure 2 shows why the interval is the primitive and the label is derived. Among the  
 175 records with four rib-free vertebrae, some reach that count through a lumbar rib and some  
 176 without one; the two are indistinguishable by the count alone and are distinguished here  
 177 only because rib presence is itself labeled.

178 Sixteen records carry a lumbar rib in the released volumes: thirteen bilateral and three  
 179 unilateral, all three on the right. That laterality is reported because it is what the release  
 180 contains and not as a finding — sixteen records cannot support a statement about side  
 181 preference, and none is made.

182 Eighteen records carry an L6. Fourteen are labeled lumbarization, two sacralization,  
 183 one semi-sacralization and one normal, which is worth stating because it shows the two  
 184 annotation axes disagreeing in the open: a record can carry six lumbar-type vertebrae and  
 185 still be read as a sacralization, or as unremarkable, depending on where the reader started  
 186 the count.

Both counts are derived from the released label volumes, and so are the manifest fields that report them. That is a correction rather than a design note: through v5 the `has_l6` flag was true for exactly one record, which contains no L6, and false for all eighteen that do, while `has_lumbar_rib` was false everywhere and `n_lumbar_labels` read zero in 799 of 802 records. A reader selecting the six-lumbar cases on those fields would have received one record that is not one, and a reader selecting lumbar ribs an empty set, with nothing to indicate either. The fields are rebuilt for this release by counting identifiers 25 and 74–75 in each volume, and `lumbar_rib_side` is added alongside them.

## II.F. Computational tools

*Access.* Every processing step described above is performed by code released under an open-source licence and archived with the dataset; no step depends on software that is not publicly obtainable. The release archive carries the reference loader, the label-scheme definition and the quality-control scripts that generated the tables in this manuscript.

*Versions and parameters.* Segmentation of femora, the sacral sub-division and the per-level rib numbering used TotalSegmentator<sup>11</sup> (`total` task, release 2.x) on a single graphics processing unit. The binary rib network is Möller’s<sup>10</sup> released weights, applied through the nnU-Net v2 framework<sup>13</sup>. The pelvic completion for records whose pelvis was absent used a five-fold nnU-Net v2 ensemble, applied out-of-fold: each record is completed by the fold that never trained on it, so no record is completed by a model that has seen it. Image handling throughout used NiBabel and SciPy; no image is resampled or reoriented when a label is written, so every label shares its image’s grid and affine exactly.

*Generative artificial intelligence.* Large-language-model assistance was used for manuscript preparation — drafting and editing of text, and generation of analysis and figure code — under author review. It was not used to generate, annotate or interpret any imaging data, and no scientific claim in this manuscript originates from it. All numbers reported here are computed by the released code from the released volumes.

TABLE I Release validation. Gates run in order and a failure stops the chain.

| Check                                             | Result                  |
|---------------------------------------------------|-------------------------|
| Geometric invariants (all cases)                  | 802/802 pass            |
| label shape, affine, spacing match CT             | —                       |
| no identifier outside the scheme                  | —                       |
| left/right structures correctly sided             | —                       |
| Ribs present                                      | 11,548                  |
| not evaluable (expected vertebra outside the FOV) | 5,788                   |
| no vertebra within the anchor distance            | 11                      |
| <b>evaluable</b>                                  | <b>5,749</b>            |
| matching the expected vertebra                    | 5,747                   |
| residual offset                                   | 2 (0.035% of evaluable) |
| misnumbered cases                                 | 0                       |
| Spinopelvic medians within published range        | 6/6                     |

## II.G. Validation

Validation is layered, and the ordering matters: measurements computed on a corpus that has not established internal consistency do not look broken, they look finished.

*Geometric invariants.* Every case is checked for label geometry agreeing with its CT in shape, affine and voxel spacing; for identifiers within the published scheme; for a non-empty label; and for sided structures falling on the correct sides, with the left–right axis read from the affine rather than assumed. The check exists because a renumbering pass once wrote a reoriented array under a canonical affine and transposed a label away from its image, a failure invisible downstream.

The check compares *each sided pair separately*, and both simpler alternatives miss cases it catches. Testing the ribs alone passed all 802 records while four of them carried `left_hip` on the patient’s right. Pooling all sided structures into one test recovered three of those four and missed the fourth, because hips and femora outweigh the ribs by voxel count, so a large correctly-sided structure masks a smaller transposed one. A gate that passes everything is not evidence of a clean corpus until it has been shown capable of failing.

228 *Rib-vertebra incidence.* Each rib is matched to its nearest vertebra and compared against  
 229 the vertebra its number implies; the chain is given in Table I. The denominator is stated  
 230 there rather than only the numerator because quoting the two residual offsets against all  
 231 11,548 ribs would halve the rate by counting ribs the check never examined — and because  
 232 more than half the ribs in a screening abdominal CT being unevaluable is itself a fact  
 233 about the corpus worth reporting. Both offsets are field-of-view truncations on individually  
 234 characterized cases, one at +1 level and one at −1, and are reported rather than suppressed:  
 235 a threshold tuned until the count reaches zero yields a cleaner number and a less informative  
 236 one.

237 The same check carries an invariant that reads as a null result. Of the 5,749 evaluable  
 238 ribs, *zero* are assigned to a lumbar vertebra — not because no lumbar ribs exist, since 16  
 239 records carry one, but because such a rib takes class 74 or 75 and never enters the numbered  
 240 set the test examines. A numbered rib landing on a lumbar vertebra would be a counting  
 241 error of exactly the kind this dataset exists to expose.

242 *Agreement with published values.* Derived spinopelvic measures are compared against  
 243 Vialle *et al.*<sup>14</sup> (Table II). Pelvic incidence is the strongest of these checks because it is a  
 244 morphological property of the pelvis rather than a posture, and so is directly comparable to  
 245 a standing reference cohort.

## 246 II.H. Castellvi typing

247 Every record whose vertebral count departs from five carries a Castellvi grade<sup>15</sup> in the  
 248 released manifest: 33 cases, typed I–IV with the *a/b* unilateral–bilateral qualifier. This is  
 249 released as an annotation layer because the grade and the count are *different axes* and the  
 250 dataset is built to show that they are: grade IIb occurs here at rib-free counts of four, five  
 251 and six alike, and seven of the 33 graded cases carry a perfectly normal count of five. A  
 252 corpus that recorded only the count would describe those seven as unremarkable.

253 The grades were assigned by a radiology resident physician. The distinction the classifi-  
 254 cation turns on — bony fusion against articulation without it<sup>2</sup> — is the one carrying clinical  
 255 weight.

TABLE II Derived measures against a published asymptomatic reference ( $n = 260$ , standing)<sup>14</sup>. This cohort is supine, and the two postural measures depart in opposite directions — sacral slope  $4.1^\circ$  lower, pelvic tilt  $4.3^\circ$  higher. That is one discrepancy, not two:  $PI = PT + SS$  holds by construction, so with pelvic incidence at its published value any fall in one appears as an equal rise in the other.

| Measure                  | This dataset (supine) | Reference (standing)     |
|--------------------------|-----------------------|--------------------------|
| Pelvic incidence         | $54.7^\circ$          | $54.7 \pm 10.6$          |
| Pelvic tilt              | $17.3^\circ$          | $13 \pm 6$               |
| Sacral slope             | $36.9^\circ$          | $41.0 \pm 8.4$           |
| Lumbar lordosis (supine) | $52.7^\circ$          | $43 \pm 11.2$ (standing) |
| PI – LL mismatch         | $2.6^\circ$           | centered near 0          |

## 256 II.I. Surgical hardware, and why a threshold is not enough

257 Metal is trivial to detect and hard to interpret, and the difference matters here for one rea-  
258 son: **an iatrogenic fusion is indistinguishable from a congenital one to a distance**  
259 **measurement**. A cage-bridged interspace reads as “no gap” exactly as a congenitally fused  
260 transitional vertebra does, so an unrecorded instrumented case is a false positive on the cen-  
261 tral measure — and identifiers 76–79 were declared in every prior version and populated in  
262 none.

263 *II.I.0.a. Detection over-calls by a factor of seven.* A threshold at 1800 HU inside a shell  
264 around the labeled skeleton flags 84 of the 802 records. Raising it to 2500 HU — the lower of  
265 the two values validated in the metal-segmentation literature — leaves 52 with a component  
266 above a 40-voxel floor. The same radiology resident physician read all 52 and confirmed 11,  
267 rejecting 41 (79%) as contrast, calcification and reconstruction artefact. Instrumentation  
268 prevalence in this cohort is 1.4%, not the 10.5% an unreviewed threshold reports.

269 *II.I.0.b. Saturation does not separate implant from artefact.* Both groups reach the scan-  
270 ner ceiling and 4 rejected proposals *exceed* it, peaking at 11,798 HU — the signature of  
271 reconstruction overshoot rather than of denser metal. What separates them is position and  
272 size together: a real surgical site *and* volume, every confirmed implant at  $2,586 \text{ mm}^3$  or  
273 above and every rejection at  $1,768 \text{ mm}^3$  or below.



274 *II.I.0.c. The subtype block had to be extended.* Identifiers 76–79 name spinal instrumen-  
 275 tation — generic, cage, screw-and-rod, plate — and cannot describe what this cohort holds,  
 276 so **80 arthroplasty**, **81 sacroiliac screw** and **82 osteosynthesis** were added. The dis-  
 277 tinction is clinical, not cosmetic: osteosynthesis holds parts of the same bone together where  
 278 arthroplasty replaces a joint, and without it the shape rules call a femoral stem a rod, since  
 279 a stem is long and thin and that is all “linear” tests. The confirmed cases comprise 8 arthro-  
 280 plasties, one osteosynthesis, one sacroiliac screw fixation and one pair of threaded cylindrical  
 281 interbody cages.

282 *II.I.0.d. Metal outranks bone, and that is a subtraction.* In every confirmed case the  
 283 implant was already inside a bone label, a segmenter handed a bright object against a  
 284 cortical surface having taken it for bone. Naming the hardware therefore reclaims voxels  
 285 rather than adding them: 1,538,852 voxels across the eleven records. The consequence  
 286 reaches past bookkeeping. Pelvic incidence and pelvic tilt are measured from the femoral  
 287 head center, and in 8 of these cases that head is an implant — any spinopelvic parameter  
 288 previously computed from them was measured on metal.

### 289 III. DATA FORMAT AND USAGE NOTES

290 Each record is a gzipped NIfTI image/label pair sharing an identical  $4 \times 4$  affine, canon-  
 291 icalized to PIR, requiring no resampling. Masks are NIfTI rather than DICOM: the policy  
 292 prefers DICOM where applicable, and for volumetric segmentation masks NIfTI is what the  
 293 downstream tooling expects.

294 Splits are patient-grouped and LSTV-stratified five-fold, frozen and shipped with the  
 295 data as `splits_5fold.json`. The five validation folds partition all 802 records, so the  
 296 release provides cross-validation and *not* a held-out test set. Users reporting a single held-  
 297 out number should carve one out and state which records it holds, because the folds shipped  
 298 here will not do that job.

299 Stratification is on the transitional subtype rather than on a binary LSTV flag, and it  
 300 enforces a minimum of each rare subtype in every validation fold: each fold’s validation  
 301 set carries three sacralization and three lumbarization records, so no fold is evaluated on a  
 302 cohort missing the class the dataset exists for. With 15 lumbarization and 15 sacralization  
 303 records across 802, that is the practical limit of what stratification can guarantee, and it is

TABLE III Key data types and metadata, and the fraction of the 802 released records for which each is populated. Counts are obtained by reading the released manifest, not by assertion. A field below 100% is not an error: age and sex are absent where the source collection did not publish them, a pelvic series identifier exists only where a pelvic annotation was placed, and a Castellvi grade exists only for the transitional records that were graded.

| Field                                                         | Records | Available |
|---------------------------------------------------------------|---------|-----------|
| Identifier, image path, label path, configuration, match type | 802     | 100.0%    |
| Patient position, tube potential, section thickness, kernel   | 802     | 100.0%    |
| Transitional label and class                                  | 802     | 100.0%    |
| Spine and pelvic annotation origin                            | 802     | 100.0%    |
| Instrumentation and partial-annotation flags                  | 802     | 100.0%    |
| Image/label alignment check                                   | 802     | 100.0%    |
| Spine series identifier; bone-fraction check                  | 782     | 97.5%     |
| Scanner manufacturer and model                                | 768     | 95.8%     |
| Sex                                                           | 749     | 93.4%     |
| Age                                                           | 709     | 88.4%     |
| Pelvic series identifier                                      | 713     | 88.9%     |
| Castellvi grade                                               | 33      | 4.1%      |

304 why a fold-level metric on the rare classes carries an error bar far wider than its own decimal  
305 places.

306 Table III lists the key data types and metadata against the fraction of records for which  
307 each is populated. Two entries are worth reading carefully. The Castellvi grade is present  
308 for 4.1% of records because it is defined only for the transitional ones; among those it is  
309 complete. A pelvic series identifier is present for 88.9% because the crosswalk records it  
310 only where a pelvic annotation was actually placed, and users joining on it should expect  
311 the shortfall rather than treat it as loss.

312 The archive of record is Zenodo (10.5281/zenodo.22139642). Code and documentation  
313 are on GitHub at the tagged release commit; any model-hub copy is a convenience mirror,  
314 not the archive.

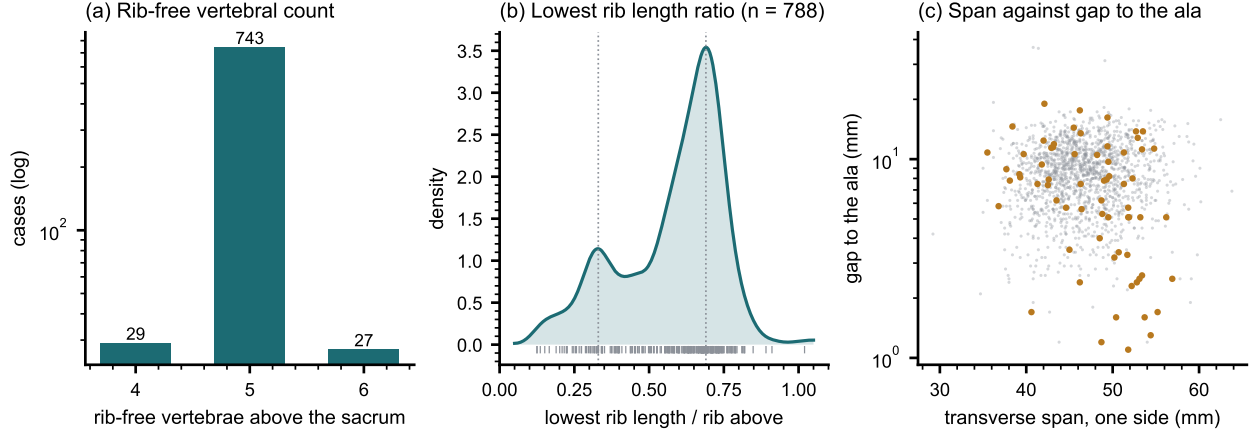


FIG. 3 Count-free measures, valid without knowing which vertebra is which. (a) Vertebrae between the lowest rib and the sacrum: an interval, not a level assignment. (b) Lowest rib length as a fraction of the rib above, showing two populations near 0.33 and 0.69 rather than one distribution with a tail. (c) Lowest lumbar transverse span against its gap to the sacral ala; cases carrying a source transitional label are marked.

#### IV. DESCRIPTIVE ANALYSIS

The cohort is a colorectal cancer screening population of 802 records. 738 are recorded female (393) or male (345); 11 carry DICOM’s *other* value, a recorded value and not a missing one, and 53 carry no sex field. Age is present for 709 (median 59, range 50–89). All 802 are counted in totals, and records outside the female/male dichotomy are excluded from sex-stratified measures — a limitation of those measures rather than of the cohort, stated because folding eleven people into “unrecorded” would misdescribe the source. The lower age bound is the screening guideline rather than a selection applied here, and it is why the measures below sit closer to a reference range than a surgical series would.

*Count-free measures* (Fig. 3). Five rib-free vertebrae above the sacrum is typical; four and six are where transitional anatomy sits, and which of the two is present does not by itself determine what to call it. The lowest-rib length ratio is bimodal.

*Level gradients and spinopelvic measures* (Fig. 4). Vertebral dimensions increase caudally in step with the load each level carries. Pelvic incidence does not vary with age, while pelvic tilt increases — the descriptive signature of a pelvis retroverting as a spine ages.

*Opportunistic measures*. Every scan carries a vertebral bone density measurement at no additional dose.<sup>16</sup> L1 trabecular attenuation is reported at the published standard site.

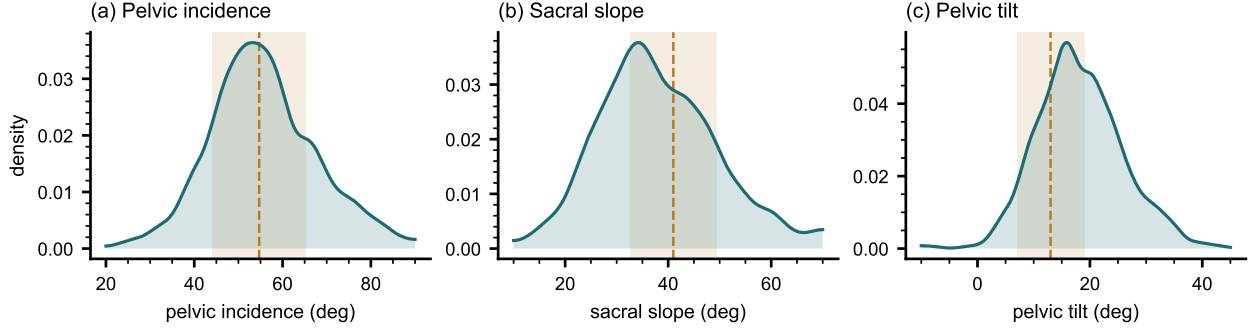


FIG. 4 Spinopelvic measures against published reference bands ( $\text{mean} \pm \text{SD}$ ) from an asymptomatic standing cohort.<sup>14</sup> Sacral slope is expected to sit below a standing reference because it is postural and this cohort is supine; pelvic incidence is not postural, which is why it can be compared without apology and why its agreement is the strongest check here.

## V. POTENTIAL APPLICATIONS AND LIMITATIONS

The principal use is the *spinopelvic interface*: spine and pelvis carry radiologist-sourced annotation on one coordinate frame in 802 records, which makes sacral morphology measurable against the lumbar column rather than in isolation — the sacral endplate and its slope, the alar corridors constraining S2-alar-iliac and iliac screw trajectories, and the L5–S1 geometry a lumbosacral fusion must cross. Pelvic incidence is derived from the segmented sacrum and femoral heads rather than from landmarks placed on a radiograph, so it is reproducible from the labels themselves.

The transitional layer serves that use rather than competing with it: no trajectory can be planned across a junction whose segments cannot be named. Level-numbering benchmarks, variant studies and opportunistic screening follow from the same annotation.

### V.A. Three uses the released measurements support

*Naming a level from its own geometry.* Wrong-level spine surgery runs at roughly one in 3,110 spinal procedures, and the cause most often cited is the transitional anatomy this cohort was assembled around.<sup>17</sup> The failure is structural rather than careless: the count is ambiguous exactly where the variant sits. Per-level body height, canal width, endplate width, transverse-process span and wedge ratio are released for T11–L5 across all 802 records, and because the labels were assigned against the twelfth rib rather than by enumeration, a

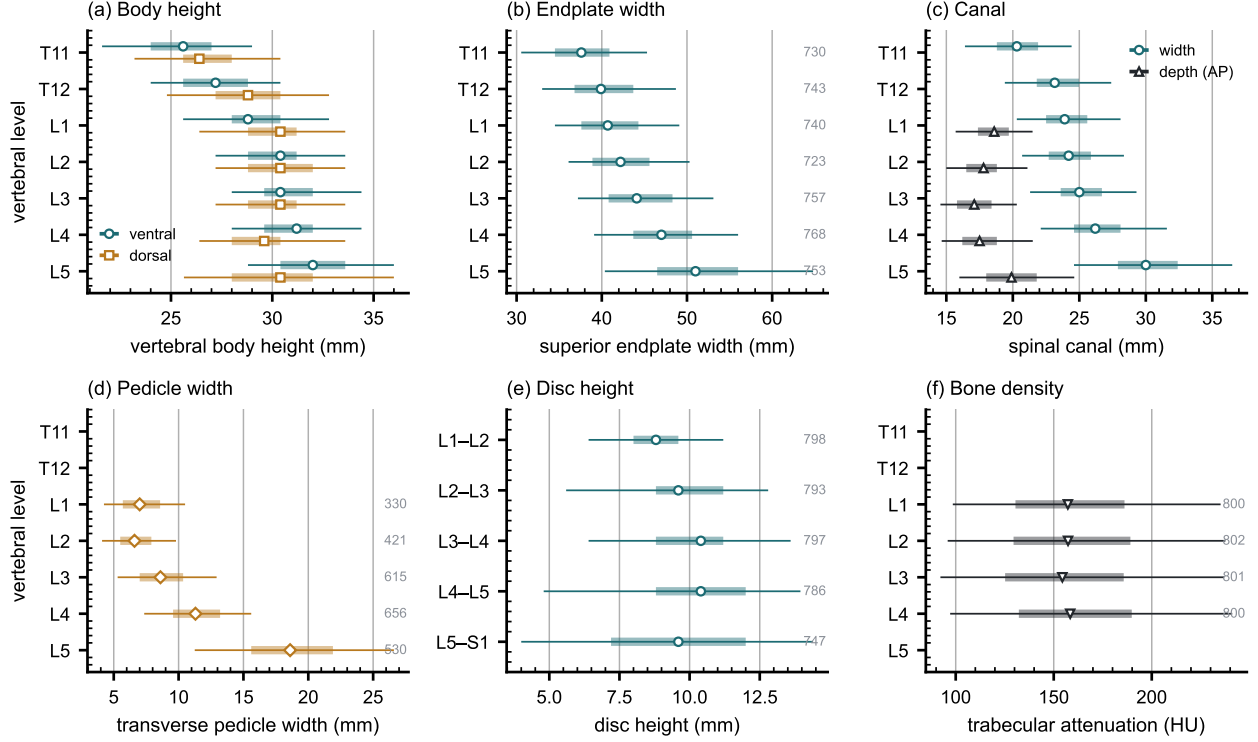


FIG. 5 Lumbar and lower thoracic morphometry by level, in the format the standard reference figures use — level on the vertical axis, measurement on the horizontal. Marker, median; bar, interquartile range; line, 5th–95th percentile; the number at the right of a row is its  $n$ . (a) Vertebral body height, ventral against dorsal. (b) Superior endplate width. (c) Canal width against depth. (d) Transverse pedicle width. (e) Disc height, drawn at the interspace it occupies. (f) Trabecular attenuation at the published opportunistic-screening site. Coverage is not uniform and is drawn rather than assumed: canal depth and pedicle width rest on 330–656 records because they need an axial extent many field-limited series do not carry, while the remaining measures rest on 720–784. No comparison between levels is tested and none is implied; the intervals are dispersion.

350 model trained on them is not learning the convention that fails.

351 *Reference morphometry with its spread.* The values a surgeon consults for vertebral  
 352 body height, canal dimensions and pedicle width come from cadaveric series of a few dozen  
 353 specimens, plotted as single values with no indication of spread.<sup>18</sup> They cannot tell a reader  
 354 whether the patient in front of them is unusual, because they never show what usual looks  
 355 like. The same measures are given here from 802 records with the distribution attached  
 356 (Fig. 5), reproducing two properties of the classical figures — body height crosses over,

dorsal exceeding ventral at T11–T12 and reversing by L4–L5, and pedicle width widens steeply into the lower lumbar spine — and adding the width of each distribution, which is the part needed to judge an individual. A larger  $n$  sharpens a distribution without changing the population it came from, and this one is a screening cohort aged 50 and over.

*Patient-specific models.* Planning is moving toward patient-specific biomechanical models rather than population norms.<sup>19</sup> Such a model needs spine, pelvis and femoral heads in one frame, which is this release’s construction; in 351 patients the annotation sits on two acquisitions in different positions, giving a within-patient change in alignment against which a predicted postural response can be checked.

Its limitations are part of it, and they are stated at the level of detail a reader would need to catch them independently.

*Coverage.* Thoracic ground truth is field-of-view limited and does not extend to T1. Postural angles are supine; pelvic incidence is not postural and needs no such caveat. The cohort is a colorectal screening population aged 50 and over, so its distributions should not be read as representative of a surgical one.

*Declared but empty, and confirmed so.* The soft-tissue identifiers (58–73) and the partial-annotation sentinel (255) are declared in the scheme and occur in no record — established by counting identifiers across all 802 released volumes rather than asserted. Both are retained so a later release can use them without renumbering, and their absence should be read as absence of annotation rather than absence of the structure. The hardware identifiers were in this position until v6 and are no longer: 76–82 are populated in the eleven instrumented records described in Sec. II.I.

*Strength of the labels varies by structure, and the release does not average over that.* Vertebral labels derive from radiologist-supervised source annotations, corrected where those sources were wrong. Pelvic labels on records that lacked one are pseudolabeled. The rib layer is a pseudolabel whose human review was triaged by an automated rule rather than exhaustive, so its guarantee is the absence of the failure modes that rule detects. Transitional labels derive from two independent sources and are not uniformly adjudicated — which is precisely why the measures reported here are count-free — and the Castellvi grades were assigned by a radiology resident physician. The sacral sub-division inherits its level from an automatic method, and in 100 records that level places S1 above half the sacrum or below fifteen percent of it, which no first sacral segment is; those records are flagged in the released

quality-control table and should be filtered before any measurement depending on the sacral endplate.

*Structures are not universally present.* The same census shows the release is not uniformly complete. Sacrum, hips and femora are present in all 802 records. One record carries no S1, one lacks T12, and nine lack an L5 identifier — in every case because the structure lies outside the field of view or, for L5, because the lowest lumbar segment is labeled L6 or incorporated into the sacrum. Users filtering on the presence of the caudal anchor should filter explicitly rather than assume it.

## VI. DISCUSSION

*Comparison with related material.* Against the collections in Fig. 1, the contribution is the crosswalk placing spine and pelvis on the same series, together with classes for the anatomy an enumeration anomaly actually produces: where another scheme must record a sixth lumbar vertebra or a lumbar-borne rib as something else, this release records it as itself.

## VII. CONCLUSION

CTSpinoPelvic1K places spine, pelvis, per-level ribs and femora on one coordinate frame in 802 records, and gives the anatomy that makes lumbar numbering ambiguous a class of its own rather than recording it as something it is not. Because both counting anchors are explicit, a level can be identified from a field of view that cannot support the conventional count downward from C2 — which is every record here. The release is archived under a persistent identifier, ships what is needed to verify it, and states per structure how strong each label is.

## DATA AVAILABILITY

The release described here is v7, permanently archived at <https://doi.org/10.5281/zenodo.22242745>. The dataset as a whole, across versions, is cited as <https://doi.org/10.5281/zenodo.22139642>, which always resolves to the current version. The archive carries the reference loader, the label-scheme definition and the quality-control tables, and the

analysis code is distributed with it under an open-source licence. The repository URL is omitted here because it identifies the authors; it is given on the title page and will be restored to this section for the camera-ready version.

## CONFLICT OF INTEREST

The authors have no relevant conflicts of interest to disclose.

## ETHICS

This work uses publicly available, de-identified imaging from The Cancer Imaging Archive<sup>8,9</sup> and annotations derived from it. The investigators obtain no identifiable private information and have no interaction with the individuals imaged, so the activity does not meet the definition of human participant research under 45 CFR 46.102. A determination to that effect has been submitted to the authors' institutional Review Board.

## REFERENCES

- <sup>1</sup>J. Lian, N. Levine, and W. Cho, A review of lumbosacral transitional vertebrae and associated vertebral numeration, *Eur. Spine J.* **27**, 995–1004 (2018), doi:10.1007/s00586-018-5554-8.
- <sup>2</sup>G. P. Konin and D. M. Walz, Lumbosacral transitional vertebrae: classification, imaging findings, and clinical relevance, *AJNR Am. J. Neuroradiol.* **31**, 1778–1786 (2010), doi:10.3174/ajnr.A2036.
- <sup>3</sup>Y. Deng et al., CTSpine1K: a large-scale dataset for spinal vertebrae segmentation in computed tomography, *Mach. Learn. Biomed. Imaging* **3**, 824–832 (2025), doi:10.59275/j.melba.2025-gf84.
- <sup>4</sup>P. Liu et al., Deep learning to segment pelvic bones: large-scale CT datasets and baseline models, *Int. J. Comput. Assist. Radiol. Surg.* **16**, 749–756 (2021), doi:10.1007/s11548-021-02363-8.
- <sup>5</sup>A. Sekuboyina et al., VerSe: a vertebrae labelling and segmentation benchmark for multi-detector CT images, *Med. Image Anal.* **73**, 102166 (2021), doi:10.1016/j.media.2021.102166.



<sup>6</sup>H. Liebl et al., A computed tomography vertebral segmentation dataset with anatomical variations and multi-vendor scanner data, *Sci. Data* **8**, 284 (2021), doi:10.1038/s41597-021-01060-0.

<sup>7</sup>L. Jin et al., RibSeg v2: a large-scale benchmark for rib labeling and anatomical centerline extraction, *IEEE Trans. Med. Imaging* **43**, 570–581 (2024), doi:10.1109/TMI.2023.3313627.

<sup>8</sup>K. Smith et al., Data from CT COLONOGRAPHY, The Cancer Imaging Archive, 2015, doi:10.7937/K9/TCIA.2015.NWTESAY1.

<sup>9</sup>K. Clark et al., The Cancer Imaging Archive (TCIA): maintaining and operating a public information repository, *J. Digit. Imaging* **26**, 1045–1057 (2013), doi:10.1007/s10278-013-9622-7.

<sup>10</sup>H. Möller et al., Automated thoracolumbar stump rib detection and analysis in a large CT cohort, 2025, arXiv:2505.05004. doi:10.48550/arXiv.2505.05004.

<sup>11</sup>J. Wasserthal et al., TotalSegmentator: robust segmentation of 104 anatomic structures in CT images, *Radiol. Artif. Intell.* **5**, e230024 (2023), doi:10.1148/ryai.230024.

<sup>12</sup>A. Schehr, J. Kim, and G. Schwing, OpenSpineConsortium: an open-source framework for medical student engagement in computational spine imaging research, *Cureus* (2026), Published 14 July 2026. doi:10.7759/cureus.112661.

<sup>13</sup>F. Isensee, P. F. Jaeger, S. A. A. Kohl, J. Petersen, and K. H. Maier-Hein, nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation, *Nat. Methods* **18**, 203–211 (2021), doi:10.1038/s41592-020-01008-z.

<sup>14</sup>R. Vialle, N. Levassor, L. Rillardon, A. Templier, W. Skalli, and P. Guigui, Radiographic analysis of the sagittal alignment and balance of the spine in asymptomatic subjects, *J. Bone Joint Surg. Am.* **87**, 260–267 (2005), doi:10.2106/JBJS.D.02043.

<sup>15</sup>A. E. Castellvi, L. A. Goldstein, and D. P. K. Chan, Lumbosacral transitional vertebrae and their relationship with lumbar extradural defects, *Spine (Phila. Pa. 1976)* **9**, 493–495 (1984), doi:10.1097/00007632-198407000-00014.

<sup>16</sup>P. J. Pickhardt, B. D. Pooler, T. Lauder, A. M. del Rio, R. J. Bruce, and N. Binkley, Opportunistic screening for osteoporosis using abdominal computed tomography scans obtained for other indications, *Ann. Intern. Med.* **158**, 588–595 (2013), doi:10.7326/0003-4819-158-8-201304160-00003.

- <sup>474</sup> <sup>17</sup>N. E. Epstein, A perspective on wrong level, wrong side, and wrong site spine surgery,  
<sup>475</sup> Surg. Neurol. Int. **12**, 286 (2021), doi:10.25259/SNI\_402\_2021.
- <sup>476</sup> <sup>18</sup>E. C. Benzel, *Biomechanics of Spine Stabilization*, Thieme, New York, 3rd edition, 2015.
- <sup>477</sup> <sup>19</sup>E. Beaulieu et al., A review of finite element analysis in spine surgery decision-making, J.  
<sup>478</sup> Clin. Med. **15**, 2584 (2026), doi:10.3390/jcm15072584.

## Supplementary Material

CTSpinoPelvic1K: spine, pelvis, ribs and femora in one coordinate frame, annotated for lumbosacral transitional anatomy

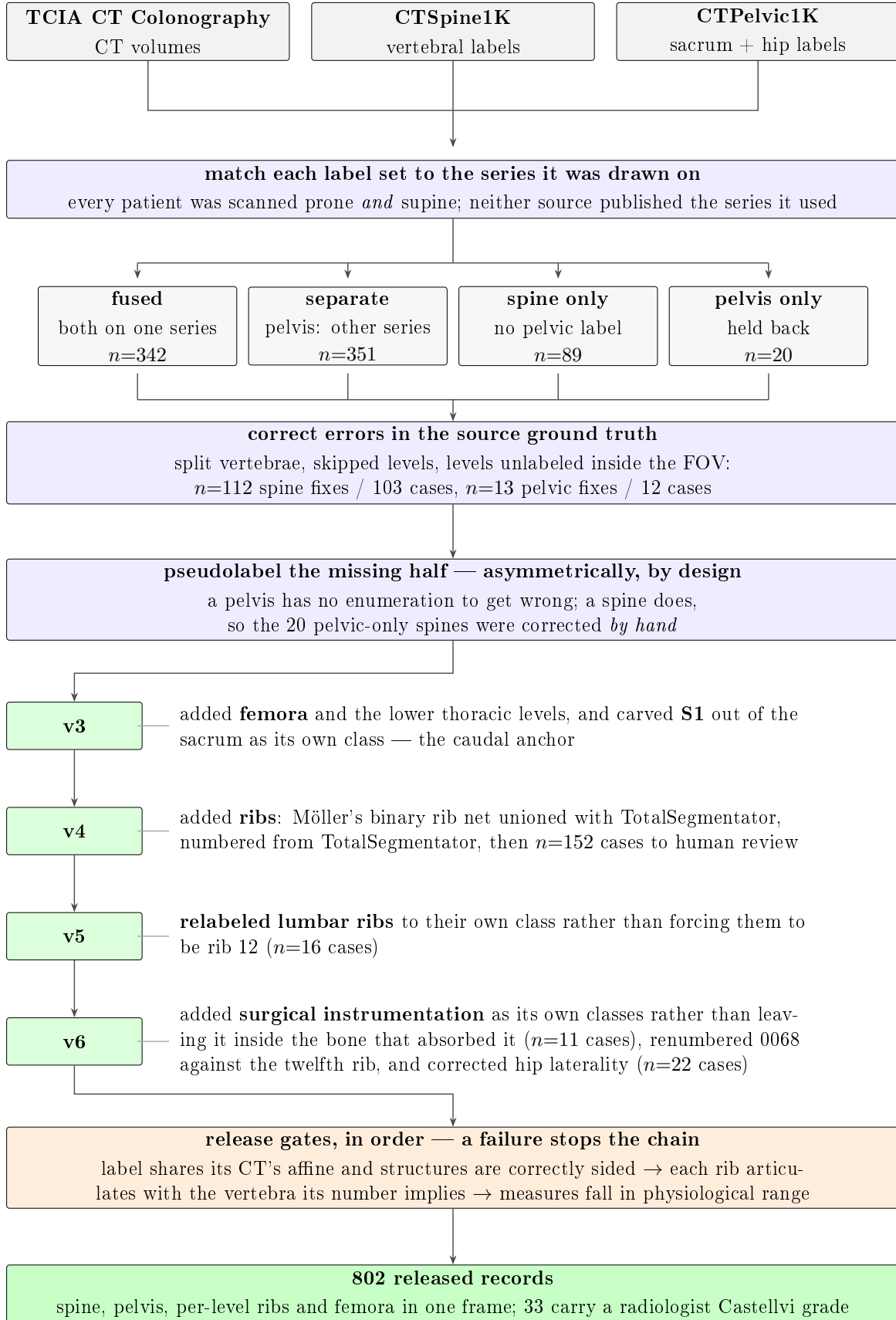


Figure S 1: How the dataset was built. Three public collections are combined patient by patient. Neither annotation source published the CT series its labels were drawn on, so each label set is matched to the series whose bone agrees with it. In 351 of those patients the two collections had drawn their outlines on *different scans of the same person*. Numbers marked *n* are scans, or corrections, at that step.